CHAPTER

A Chapter-Level Note with a Deliberately Long Explanatory Title

Geometry becomes clearer when the structure can breathe.

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Contents

1 Published chapter material (Tag `0CH01`)

Theorem 1.1 (A small test theorem) (Tag `0CH0101`). A quiet page can retain a strong hierarchy and a permanent identity.

1.1 A tagged subsection (Tag `0CH0102`)

The compact references are Section 1 (Tag `0CH01`), Subsection 1.1 (Tag `0CH0102`), and Theorem 1.1 (Tag `0CH0101`).

2 Draft environment gallery (Draft tag `0CH02`)

2.1 Statements and proofs (Draft tag `0CH0201`)

Slogan *A template should make the mathematical hierarchy visible at a glance.*

Theorem A (Chapter comparison). The three compilation modes share one statement vocabulary.

Definition 2.1 (Affine spectrum). For a ring A, the space $\mathrm{Spec}(A)$ consists of its prime ideals with the Zariski topology and its structure sheaf.

Proposition 2.2. The inverse image of a basic open set under a morphism of affine schemes is basic open.

Lemma 2.3 (Localization test). Localization commutes with finite colimits of modules.

Corollary 2.4. A localization of a finitely presented module is finitely presented.

Conjecture 2.5. The comparison remains true after replacing schemes by suitable stacks.

Question 2.6 (Minimal hypotheses). Which finiteness assumption is actually used in the argument?

Example 2.7 (Draft tag `0CH0202`). The affine line is $\mathbb{A}^1_k = \mathrm{Spec}(k[t])$. The displayed draft tag is provisional and is absent from Example 2.7.

Exercise 2.8. Compute the standard-open cover of $\mathbb{P}^1_k$.

Construction 2.9 (Gluing two affine lines). Glue $\mathrm{Spec}(k[t])$ and $\mathrm{Spec}(k[t^{-1}])$ along $\mathrm{Spec}(k[t, t^{-1}])$.

Notation 2.10. Denote the resulting scheme by $\mathbb{P}^1_k$.

Remark 2.11 (About draft references). Draft references display only the current type and number.

Proof of Theorem A. Each mode installs the same public environments.

Step 1 (Compare counters). Ordinary statements share one counter; `mainthm` uses letters.

Claim 2.12. The visual style does not alter the underlying amsthm semantics.

Case 1 (Book or chapter mode). Statement numbers reset within sections.

Case 2 (Section mode). Statement numbers use a single local sequence.

□

The gallery supports references to Definition 2.1, Proposition 2.2, Lemma 2.3, and Corollary 2.4, together with Claim 2.12, Step 1, and Case 1.

2.2 Typesetting features (Draft tag `0CH0203`)

The text font is Libertinus Serif, headings use Libertinus Sans, and source-like material uses Source Code Pro. Mathematics uses Libertinus Math, with STIX Two providing distinct calligraphic and script alphabets:

$$\mathcal{O}_X(1), \qquad \mathscr{H}^i(\mathscr{F}), \qquad \mathbb{P}^n_k, \qquad \mathfrak{gl}(V), \qquad \mathbf{R}\mathrm{Hom}_X(E, F).$$

$$\begin{tikzcd} \mathrm{Spec}(B) \arrow[r] \arrow[d] & \mathrm{Spec}(A) \arrow[d] \\ Y \arrow[r] & X \end{tikzcd}$$

(a) A first chapter-level task.

(b) A task with two parts.

  (i) establish the local claim;

  (ii) glue the local results.

```
def sections(chapter):
    return sorted(chapter.sections, key=lambda item: item.order)
```

Listing 1: A chapter-level code example

See Code 1. The draft structural reference Section 2 omits its provisional tag.